

The background of the entire slide is a dark grey or black color, overlaid with a repeating pattern of light grey circuit board traces. These traces form a complex, interconnected grid of lines and small squares, resembling a printed circuit board (PCB) layout. The pattern is consistent across the top and bottom of the slide, framing the central text.

DIGITAL **TRANSFORMATION** **+ CHANGE** **MANAGEMENT** **IN MINING**



You Can't Have One Without the Other

GROUNDHOG
MINE DIGITIZATION & AUTOMATION

“

**THIS IS FOR CERTAIN —
YOU CANNOT SUCCESSFULLY
IMPLEMENT DIGITAL TECHNOLOGY
AND MANAGE THE IMPACTS OF
TRANSFORMING YOUR MINE
WITHOUT A PLAN**

”

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**WE ENVISIONED A TOOL THAT WAS
FOCUSED ON TRANSFORMING
THE MINING INDUSTRY**

CHAPTER ONE

INTRODUCTION

The Digital Transformation Journey

Digital transformation has been a buzzword for several years, as more and more mines seek to enhance their safety, improve production, and adapt to a changing business and social environment.

Understanding that “How might we?” is generally considered a question central to innovation by turning problem statements into solutions. The questions GroundHog posed in the beginning of this journey reflected what were, based on hundreds and maybe even thousands of conversations, the desires of every mine, everywhere in the world.

How might we safely move more tons, more efficiently, with safety being of paramount importance?

How might we get more done every shift?

How might we digitize paper-based processes and use the digitized data as an enabler of continuous improvement and reducing variation?

And finally, how might we design and develop an app, a mine digitization platform, to do all of this within a framework that makes sense for the people involved?



Figure 1.1
Surface Mine Operation

We envisioned a tool that was focused on transforming the mining industry to make things easier, enable people to get more things done that are valuable to THEM, and automate the non-value activities through smart technology. After all, everyone wants to spend more time on the actual work of production and dispense with the roadblocks and bottlenecks caused by lack of timely information and communication that inhibit what could, and should be, effective collaboration of all the moving parts that make up a mining operation.

Today, we see that the digital transformation that was considered a “nice to have” not so long ago has become an imperative – change or be left behind.

But what does digital transformation mean to your business specifically? Every tool has features, but before you jump into “transformation,” first we need to back things out to the 10,000-foot level and understand what needs to transform, and what your goals are. There is pre-planning involved even before you search out a solution that will suit your purpose.

**“ A TOOL TO MAKE
THINGS EASIER &
ENABLE PEOPLE TO
GET MORE THINGS
DONE THAT ARE
VALUABLE TO THEM**

What is your initial goal (or goals), and what are the pain points you are seeking to address?

Is it safety? Safety is always a top concern, we know that planned work is safer work, and nobody will argue that any available step change is welcomed.

What about active shift length? Perhaps alignment of planning and activity at the face? Transparency on the state of mining activity underground at any given time? Compliance to plan?

**“ YOU CANNOT
SUCCESSFULLY
IMPLEMENT DIGITAL
TECHNOLOGY
WITHOUT A PLAN**

It may be that you're looking to improve on all or some of them. But in our experience, the overarching goal of transformation should be to identify the key issues you wish to solve, and prioritize the steps required to do so into smaller "bites."

This is often where solution providers begin to differentiate from one another. To be honest, the physical implementation of the technology is the easy part. There will be many risks to consider on your way to the clear benefits of digital transformation. Where the greatest risk lies is in not using a systematic approach that respects the fact that what you are really doing is changing how people work. The last thing you want to do is overwhelm the people tasked with the responsibility of adopting a new and unfamiliar technology – especially if it is part of a bigger transformation picture.

You are on an adoption journey – it is not a “one and done” process. You don't have to do it all at once, which is why when looking for the solution that is the right “fit”, you want to know that you are working with someone who can help guide you with an established and proven digital transformation roadmap. As you will see later, this is key to GroundHog's approach.

One this is for certain – you cannot successfully implement digital technology and manage the impacts of transforming your mine without a plan for how you will manage all aspects of the change, from the specific project to the tech itself, the impacts on other stakeholders, and most importantly, on the people expected to do the work, day to day.

Don't forget...

Your digital transformation probably isn't the only project going on! What else is taking place concurrently with your plans?

- ☐ BATTERY ELECTRIC VEHICLES
- ☐ NEW SAFETY SYSTEMS
- ☐ EQUIPMENT TRIALS
- ☐ NEW CONSTRUCTION
- ☐ IT INFRASTRUCTURE IMPROVEMENTS
- ☐ ENVIRONMENTAL INITIATIVES

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**THE SECOND CRITICAL ASPECT OF
ANY TRANSFORMATION INITIATIVE —
MANAGING THE PEOPLE**

CHAPTER TWO

PEOPLE, PROCESS, TECHNOLOGY

Managing All Aspects of Change

From experience, we can happily confirm that with the right commitment to managing all aspects of change through an established roadmap, the potential for successful outcomes is vastly improved. Methodically moving through the levels offers a smoother transition from existing systems to a digital, shift-based planning tool, based on a People, Process, Technology philosophy. We will discuss what this means in more detail later.

Suffice it to say that when one starts down this road, the findings are usually that the technology works, and the production increases occur, but despite encouraging returns on digital implementation, there is resistance. People are frustrated and unhappy, and end up blaming the tech.

This is where we arrive at the second critical aspect of any transformation initiative – managing the people side of the change. What must be understood is the magnitude of change management, the amount of planning required, and that implementation does not equal adoption. If the assumption is that people will figure it out, that it should be easy, the challenge in front of you will be more challenging than it needs to be, and in fact, with no focus on managing change, the long-term success of ALL your transformation efforts may be in question.



Figure 2.1.
*Utilizing Collected
Data to Run the Shift*

Why? To use the old aphorism...once bitten, twice shy.

A new mine manager had just been installed by head office to lead a digital initiative to collect data that would provide decision support. His reputation within the industry was excellent, he was well-respected by workers and management alike, and nobody doubted he would get the job done – until he encountered significant resistance from the outset. One of the main complaints? That the initiative he was leading was just the “flavor of the week.”

He soon found out that several changes had been pushed down the line by “corporate” with high expectations but little regard for how it affected production or the workers during the learning curve. Beyond the expectations of “success,” little communication of any rationale as to why the changes were even necessary, never mind how they could be useful to their work or benefit them, had taken place. As one after another initiative was either partially adopted or not adopted at all, with no enthusiasm or champions of change, and nobody who could explain why the new processes or technologies were “better,” frontline managers and workers became jaded to new change initiatives. The project soon became another good idea with poor execution and no long-term benefit.

When looking at why past change initiatives didn’t have the desired uptake, start with what communication was there from leadership, middle management, and whether the people understood what was in it for them.

“ YOU WILL WANT A DATA DRIVEN MODEL FOR SOFTWARE ADOPTION

At risk of being repetitive, your focus should be on the:

People. Then...

Process. Then...

Technology.

This means that for a successful implementation, a deep understanding of the human reaction to change is required. You need a process that makes sense to the way they work. Then, and only then, do you start to focus on the technology.

You will want a data driven model for software adoption. And, of course, the right tool for your mine. But beginning to manage your anticipated transformation through effective change management should begin long before the project phase. In fact, if you are reading this – it starts now.

Opinions differ on what change management even means. For example, people with a project management background may focus on tools and processes used to manage change within a project and its team. But these are typically more focused on reacting to changes that may occur to a project plan.

Of course there should also be a clear vision for the expected results, and ensuring data can be collected and validated so there is ongoing reporting and you are able to immediately react to challenges and reinforce the successes.

But for the purposes of this discussion, change management focuses on the people side of change. We'll get a bit more detailed later, but in general, change management is designed to help you assess the risks, effectively assess and communicate the need(s), and understand and communicate the anticipated benefits and potential impacts to all stakeholders, at all phases of the change.

What is important to the people involved? How might it affect their roles and responsibilities? Their incomes? We're not talking about an academic or theoretical take on change – work represents peoples' livelihoods, which directly impacts every facet of their lives. It's important stuff. This is why from the conception of the need to change the way your people work, to well beyond the end of the project, change management plays a key role in the success of your transformation efforts, and to the success of your business overall. It truly cannot be understated.

“ CHANGE MANAGEMENT RELIES ON INTELLIGENCE GATHERING

At its core, change management relies on intelligence gathering. There are several assessments that can be done at every phase that will guide you towards success. A good change management consultant is not unlike a detective, asking the right questions at the right time can help avoid a lot of wasted effort and resources, and the feedback you receive can point you down the best path towards your goals.

For example, you will be able to determine what is the minimum viable adoption required, meaning the critical functionality within the tech that must be adopted to meet the vision and definition of success. (This is also where change management intersects with GroundHog's short interval control maturity model and its five levels, and you become able to determine what percentage of functionality will meet your needs.)

Beyond that, however, there are so many key bits of information:

- ☐ **WILL YOUR CULTURE SUPPORT THE EFFORT?**
- ☐ **WHAT BEHAVIORS DO YOU NEED FOR ADOPTION?**
- ☐ **WHO ARE THE INFLUENCERS IN THE ORGANIZATION?**
- ☐ **WHO NEEDS TO BE INVOLVED AND WHEN?**
- ☐ **WHAT TOOLS DO YOU USE TO COMMUNICATE?**
- ☐ **HOW RECEPTIVE IS THE AUDIENCE TO EACH ONE?**

Understanding the stakeholder groups - right down to the individual people involved in implementing your change - is not only possible, it is critical. But with the right approach, also very doable.

**UNDERSTANDING THE STAKEHOLDER GROUPS –
RIGHT DOWN TO THE INDIVIDUAL PEOPLE INVOLVED IN
IMPLEMENTING YOUR CHANGE – IS NOT ONLY POSSIBLE,
IT IS CRITICAL. BUT WITH THE RIGHT APPROACH,
ALSO VERY DOABLE.**



*Figure 2.3.
Underground Operation*

*Figure 2.2.
Engaging All Stakeholders
Through the Entire Process
Stream*

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**OFF THE SHELF COMES
OUT ON TOP EVERY TIME**

CHAPTER THREE

OFF THE SHELF OR DO IT YOURSELF?

Finding the tools and guidance

In the match up of OTS vs DIY, OTS comes out on top every time. You may be thinking, “Of COURSE they would say that! They are an OTS solution!”

But hear us out. Or better yet, read on...

A mining company in the Americas was already using a couple of different off the shelf solutions, but decided to develop its own Short Interval Control solution, with the intention of rolling it out across several sites. The business case was sound, so they moved forward with hiring dozens of talented experts for both the development and implementation, engaged consultants for the change management, and invested millions in capital expenditures and the tech they would require to roll-out their home grown SIC app. A site was chosen to pilot the solution, and the project got underway.

So they did everything right, right?

Well, there were a few gaps. A number of improvement projects taking place concurrently, and no real baseline measurement, would make it difficult to measure success. Development proved

**“ THERE ARE MANY
OTS SOLUTIONS
THAT CAN PROVIDE
YOUR MINIMUM
VIABLE SOLUTION
IMMEDIATELY**

more complicated and time consuming than was thought, and the team was beset by delays. Implementation was going well, but other tech critical to the full and effective use of the solution wasn't ready. The change management efforts were good, but demonstrating the usefulness and usability of a product that wasn't ready was challenging.

In the end, the project was shelved along with the many millions that had been poured into it, not because it failed, but because it took long enough to get off the ground that it was still all cost and no return where a management change occurred, and with no available data to support the solution with new management, the project was scuttled.

Will this happen to everyone? Probably not. Is the risk worth it? Probably not, when there are many OTS solutions that can provide your Minimum Viable Solution immediately. Most have dev teams on standby who will be responsive to your specific needs and may be able to develop desired features over time.

You don't have to do this yourself. You do not have to reinvent the wheel.



**YOU DON'T HAVE TO DO
THIS YOURSELF.**

*Figure 3.1
Teamwork Underground
Mine Operation*

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**THE BENEFITS OF USING AN
OTS SOLUTION GO BEYOND
THE TECHNOLOGY.**

CHAPTER FOUR

BEYOND TECHNOLOGY

Set Up for Success

We hope to demonstrate that the benefits of using an OTS solution go beyond the technology. For example, a good OTS solution will come with experience in implementing the product in a logical, systemic way, like our Roadmap. They will help set you up for success and ensure you have the skills to use the tool.

You can also look for a company that has experience in ensuring the implementation sticks through understanding adoption and providing the tools and guidance required. At the end of your project, a good solution provider will leave you with a repeatable and scalable process design that will help guide you through future implementations and help create a narrative – a “day in the life” picture, so to speak – that follows an arc of how you use new tech, what it replaces, and why it is better. They become a valuable ally in ensuring you are prepared with the right questions and approach for any new vendor or digital implementation. This benefits everyone and can be applied by everyone.

Through planning and process and stakeholder mapping, you should be left with an understanding of your current state, and be able to effectively control the message of who, what, when, where, why, and of course, how, when it comes to your digital transformation efforts.

When it is all said and done, there must be an understanding that your solution is more than a question of tech, with goals that go beyond ROI – ensuring adoption can dictate whether your investment remains in use, or is wasted and shelved, an outcome that is, for a host of reasons, all too common in industry.



**DIGITAL TRANSFORMATION
CAN ALSO EFFECT A CHANGE IN
COMPANY CULTURE**

CHAPTER FIVE

WHAT IS DIGITAL TRANSFORMATION?

Improve, Optimize, & Accelerate

We all love a good baseline, don't we? We don't want to make assumptions about where any company might be on their transformation arc, so before we go any further, let's establish one for what we mean when we talk about digital transformation.

Digital transformation is the integration of digital technology into some or all aspects of your business. Generally, what companies are looking for is to improve safety, optimize efficiency in process, accelerate productivity, and allow the company to deliver a greater level of value to their customers and shareholders.

But we'll say it again, the tech is the easy part. You can buy all the technology you like and install it everywhere. That does not mean you have "transformed." For a lot of mining companies, digital transformation can also effect a change in company culture that ushers in a different set of responsibilities. Its impacts can be far-reaching too - digital transformation may broaden the scope of a company's capabilities, in turn expanding objectives, employee skill sets, experimentation, and the opportunity for larger successes.

But in the vast majority of cases, it will also be accompanied by a learning curve (or curves) because it represents a change in how things operate, and to be frank, it is surprising how this escapes the attention of so many people.



Figure 5.1
Moving Tons Safer + Faster

We've note that digital transformation is critical to companies that do not want to be left behind, but what does this mean, exactly?

In the mining space, it is not so much about competition as it is in increasing value and maximizing resources. When we look around, we see the world is rapidly adopting digitization, and businesses of all sizes are realizing how important it is to embrace technology in order to retain a competitive edge in their respective marketplaces. But mining tends to be a conservative business, and therefore slower to adopt new technologies than other industries.

There are a number of reasons mining companies and/or individual sites may decide to digitize their businesses, but in our many conversations, one of the considerations is as plain as can be: survival. We all recognize the truth that the more material one can mine safely and efficiently, the greater the likelihood of success in the marketplace, ensuring the viability of the business and extending the life of mine.

And there are several advantages involved with digital transformation. Which ones would you like to see?

- ☐ **ENHANCED EMPLOYEE SAFETY**
- ☐ **INCREASED PRODUCTIVITY AND EFFICIENCY**
- ☐ **MORE EFFECTIVE USE OF RESOURCES AND EXTENDED LIFE OF MACHINERY**
- ☐ **A NEW BREADTH OF KNOWLEDGE THROUGH THE AVAILABILITY OF DATA COLLECTION**
- ☐ **THE CREATION OF NEW COMMUNICATION CHANNELS AND SHARING OF MORE INFORMATION**
- ☐ **THE CONCEPT OF TEAMWORK BEING CHAMPIONED ACROSS THE WORKFORCE**

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**PERHAPS A BETTER QUESTION IS,
“WHICH ONES CAN YOU DO WITHOUT?” AND THE
ANSWER IS...NONE OF THEM. SO THE PRE-WORK
IS IN **PRIORITIZATION**.**

An abstract pattern of light gray lines and squares, resembling a circuit board or a network diagram, located in the lower half of the page.

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**DIGITAL TRANSFORMATIONS
REQUIRE A FOUNDATION TO
BUILD UPON**

CHAPTER SIX

BUILDING BLOCKS OF DIGITAL TRANSFORMATION

Foundational Framework

No two businesses are exactly the same, so organizations will see significant variance and need in their transformations to digital based on business type, goals, and other factors. But in general, there are common factors that must be taken into account, and this commonality allows for the creation of a framework to address them. In addition to the obvious introduction of new technology, they include the need for operational readiness, creation of a culture of innovation, effective, respected, and responsive leadership, and employee enablement.

Digital transformations require a foundation to build upon and establishing effective strategies in light of these commonalities is crucial to success. Again, success is not defined by the acquisition of new digital technology, but rather by sound leadership

You are likely familiar with the concept of Plan Do Check Act (PDCA), and it fits perfectly into any roadmap to success. Planning plays a key role in digital transformation strategy, and devising a roadmap with established goals along the way is essential – quick wins demonstrate the value of the investment while you grow the use of any digital tool. An implementation roadmap broken down into two to three-month increments allows you to implement, analyze, and take action over and over. It is a journey rooted in Agile



Figure 6.1.
*Communication Within
Every Level of the Business*

and Lean thinking, not a “one and done” process. As you collect data, your decision making becomes better and easier, but only if it is captured and used at frequent daily intervals. An essential part of the journey is to ensure this becomes an entrenched part of your process.

Another benefit of creating a plan that is transparent to stakeholders across all levels, and clear in its usability and usefulness as a guide, is that it can go a long way in breaking down silos and initiating communication across operational units and departments by setting a common goal, or destination if you will.

A company’s initial financial commitment and implementation is the “easy” part, but ensuring genuine buy-in and dedication from everyone involved can be challenging, especially to make the changes “sticky” – you are looking to have your people adopt a permanent solution or process, not something that will only change for a short period of time and then be forgotten in favor of the old way of doing things, or a new initiative that shouldn’t be necessary if the current transformation project is done right the first time.

**“ ENSURING
GENUINE BUY-IN
AND DEDICATION
FROM EVERYONE
INVOLVED CAN BE
CHALLENGING.”**

While your information technology group will likely play a key supporting role in the physical implementation of the software or app in any digital transformation, and the real adaptation to the changes of any digital implementation project will inevitably fall to the supervisors and frontline workers, any change in the way you operate becomes the responsibility of everyone in the organization. This is why impact assessments are so important in seeking to understand who will be most affected. But you can be certain that the kind of digital transformation we are discussing will almost every case be more of a people management issue than simply an IT issue.

“ UNDERSTAND THEIR CONCERNS AND DETERMINE HOW TO CREATE THE DESIRE FOR CHANGE

Under any change scenario, workplace silos tend to solidify, and people may reflexively protect their “turf”. It is also common for workers to assume their value isn’t being recognized and that the way they have been doing the jobs they have become skilled at over the years is now considered “wrong,” to the extent that their jobs are in danger of being replaced by the new technology!

In collecting feedback from workers in order to understand their concerns and determine how to create the desire for change, most experts in Change Management will tell you that one of the main complaints that dogs digital implementation efforts - especially SIC efforts designed to track progress during the shift and collect that data for more effective decision making later - is that this means workers are being “watched by management.” This gives one a clue as to how the collection of this feedback, and the creation and effective communication of clear messaging about why the changes are desirable, is so important.

Of course, in reality, nobody has time to be watching individuals. But a digital transformation project of any magnitude will often re-define processes, change roles and sometimes even job titles - all changes employees tend to approach with suspicion, or even fear. This should not be downplayed or dismissed - after all, this is their livelihood.

Knowing your workforce is of key importance, as there are some specific workforce sub-populations that tend to resist change. Check off the sub-populations that you think exist in your workforce, then have a frontline supervisor do the same, and see if your results match:

- ☐ OLDER EMPLOYEES NEARING RETIREMENT
- ☐ WORKERS WHO ARE NOT TECHNOLOGY SAVVY
- ☐ LONG-STANDING EMPLOYEES WHO ARE SPECIALISTS
- ☐ HIGHLY EXPERIENCED WITH A SPECIFIC PROCESS OR PIECE OF MACHINERY
- ☐ MEMBERS OF THE WORKFORCE WHO ARE FOCUSED
- ☐ SOLELY ON ESTABLISHED PROCESS STRUCTURES
- ☐ INTROVERTED EMPLOYEES WHO PREFER TO WORK ON THEIR OWN (If the change means they must now communicate more frequently.)



Figure 6.2
Moving Tons Safer + Faster

Knowing who these individuals are can be a huge benefit, and not only so their fears can be put to rest; you'll find out soon why they can become your biggest allies.

Strong and selfless leadership is critical within these populations and demonstrating genuine empathy as a leader is essential. Leaders build trust through displaying empathy to employees who are struggling with change, in turn breaking down silos and cultivating workforce alignment. It's virtually impossible to successfully complete digital transformation when trust and support are absent.

So let's talk about why we say the tech is the easy part; you have to sell it to Lester...

On an implementation project that featured the addition of tablets in the cabs of some of the machines, one of the implementation leads from the vendor was on a ride along with "Lester", an old, crusty miner who was nearing retirement and was experienced with every piece of equipment, but loudly claimed he hated new technology and was computer illiterate. It was clear, however, that Lester had learned how to operate many new pieces of increasingly modern machinery – he was smart, just fearful that he might look "stupid" to his peers. The implementation lead took the opportunity to demonstrate one on one how the app – which fortunately was developed to be user friendly – was easy to use, and explained how the data might be used to increase tonnage. (It is generally not advised to use "bonus" directly as a prime motivator,

but miners are very good at putting two and two together.) When Lester saw how easy it was to use the touch screen to enter his numbers and understood the benefit to him (including not having to keep track of a pen and piece of paper for 12 hours!), he was an immediate convert. All that was required was some guidance, encouragement, and what was in it for him.

This is why it is important to engage both official and “non-official” leaders, the latter being individuals who, by force of character, charisma, experience, or talent and skill, are looked to for guidance within their crew or work group. These individuals are generally well-known but can also be identified through a sponsor or stakeholder assessment. It is important to know who supports the change, who does not, and to what extent they are resistant or supportive – this knowledge can make or break a project. An influential detractor is a serious problem, but bringing an influential detractor on board - like Lester - can be a virtual superpower.

We will delve into what the main motivators are for the leadership levels in mining, but suffice it to say that your strongest leaders must be inspired not only to help others understand the benefits of digital transformation and why it's critical, and to motivate employees to want to make the change. If they are brought on board effectively, they will provide ownership and empowerment opportunities to employees (assuming that everyone has something to contribute). Most importantly, they'll actively listen to employees and meet them where they are, considering individual perspectives and how radical changes can make them feel. So providing them with the right messaging at the right time, and ensuring the process leaves them prepared to address questions proactively instead of reactively, is of utmost importance.

Recognizing employee sub-populations and the need to cultivate an empathetic environment before digital transformation begins is part of the importance of planning, and critical in deploying effective change management strategies.

Providing the framework, and the support through the change are truly the building blocks...so let's add a little more detail.

**IT IS
IMPORTANT
TO KNOW WHO
SUPPORTS
THE CHANGE,
WHO DOES
NOT, AND TO
WHAT EXTENT
THEY ARE
RESISTENT OR
SUPPORTIVE**



7



**THE PEOPLE EXPECTED TO USE
ANY NEW TOOL HAVE THE POWER
TO MAKE IT OR BREAK IT**

CHAPTER SEVEN

CHANGE MANAGEMENT

Systematic Transformation

As defined by Oxford Languages:

change man·age·ment
noun

1. the management of change and development within a business or similar organization.
2. the controlled identification and implementation of required changes within a computer system.

Obviously we are talking about the former definition of change management, not the latter, although we'll acknowledge that when you are talking about a digital transformation, one could be confused by the latter definition. We've already discussed how some people – particularly those with a project management background – might differ in their assessment of what change management means.

For our purposes, from the digital transformation standpoint, change management typically refers to the management of change for the people going through systematic transformation of processes, objectives, or technologies of an organization. It involves assessing for many factors that will affect adoption, implementing strategies for both controlling and bringing about change, and mainly, supporting and assisting the workforce in adapting to new ways of working.

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